# A Comparative Analysis of Text Normalization Techniques for Enhanced Sentiment Analysis Performance

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*Abstract*—**Sentiment analysis on Indonesian social media, particularly on platforms like Twitter, faces challenges due to the prevalence of informal and non-standard language. This variation complicates natural language processing (NLP) tasks, necessitating robust language normalization techniques. This study evaluates five normalization methods Jaro, Jaro Winkler, Levenshtein, Damerau-Levenshtein, and Smith-Waterman by comparing their effectiveness in enhancing sentiment classification accuracy on Indonesian Twitter data. The methods are tested against a standard word set (KBBI) and sentiment-labeled review data from the DANA application. Results indicate that Damerau-Levenshtein achieves the highest accuracy of 89.15% in classifying sentiments into positive and negative categories. Meanwhile, the Jaro method demonstrates strong consistency in handling errors like duplication and character insertion, albeit with slightly lower accuracy. These findings highlight the critical role of language normalization in improving NLP performance and provide actionable insights for developers working on sentiment analysis systems in the context of Indonesian social media.**

***Keywords—Language normalization, sentiment analysis, Grey Wolf Optimizer, Twitter Indonesia, natural language processing.***

## I. INTRODUCTION

In Indonesia, platforms like Twitter are widely used for expressing opinions and sentiments. However, the informal and non-standard language often found on social media poses challenges, making language normalization essential to improve the accuracy of sentiment analysis.

In previous research conducted in 2023, language normalization was done manually due to the limited availability of accurate automated methods, which limited the efficiency and scalability of the research [1]. Christine P. Chai in her survey on text preprocessing methods in NLP emphasized the importance of choosing the right method based on the type of data and the purpose of the application. Chai stated that text normalization, such as spell checkers, is effective in reducing spelling variations and improving model accuracy, especially for unstructured text data [2]. In 2024, Ondřej Rozinek and Jan Mareš conducted a speed and accuracy comparison between Convolutional Jaro and Jaro-Winkler Similarity, as well as other methods such as Jaro, Levenshtein, Damerau-Levenshtein, Needleman-Wunsch, and Smith-Waterman. The results showed that Convolutional Jaro produced the best accuracy and speed, followed by Jaro-Winkler for accuracy and Jaro for speed [3]. This study aims to overcome the limitations of [1] by comparing various language normalization methods to improve the accuracy of sentiment analysis on Indonesian language social media data, especially Twitter, because previously no one had compared them and evaluated their relationship with sentiment analysis, especially for Indonesian.

In this study, the normalization methods compared focus specifically on spelling error correction. Spelling errors often act as a bottleneck in text processing because they can increase the number of unrecognized words (Out-Of-Vocabulary/OOV) in the model. For example, a misspelled word like "dangudt" should be corrected to "dangdut." If this error is not addressed, the accumulation of OOV words will reduce the model's understanding of language context. Effective spelling correction, as part of the language normalization process, helps the model recognize standard words, thereby improving the accuracy of analysis, including in sentiment classification [2].

This study compares edit-based text distance algorithms such as Jaro, Jaro-Winkler, Levenshtein, Damerau-Levenshtein, and Smith-Waterman due to their ease of use without requiring model training. These methods effectively handle typographic variations directly using predetermined rules, in contrast to AI-based models that require training and large computing resources. For example, the word "ball" can produce up to 237 error variations through edit operations. With 112,651 words in KBBI, this means there are 26,698,287 possible error combinations that are challenging for AI models. Edit-based methods are more flexible and efficient because they do not require continuous dataset updates. This study evaluates these methods using KBBI data and labeled sentiment from Kaggle related to the DANA application [4], [5].

This study aims to evaluate normalization methods for correcting non-standard words and their impact on sentiment classification performance. By focusing on traditional approaches, it seeks to lay the groundwork for automatic language normalization tailored to Indonesian social media. Effective normalization can enhance sentiment analysis accuracy by improving the model's understanding of informal language. The findings will offer practical recommendations for developers and researchers to build reliable systems capable of recognizing non-standard language variations and accurately interpreting user sentiments.

## II. Literature Review

### A. Language Normalization

In this study, a comparison of several normalization methods was conducted to determine which one is most suitable for sentiment analysis. The normalization techniques used are as follows:

#### 1) Jaro

This method measures the similarity between two strings by considering the matching characters and the number of transpositions required. Jaro's method is effective in handling minimal character errors, but has limitations when dealing with words that require many changes [3], [4]. The formula of Jaro is shown in (1).

$$S_J(S1, S2) = \begin{cases} 0 & if\ m = 0, \\ \frac{1}{3}\left(\frac{m}{[S1]} + \frac{m}{[S2]} + \frac{m-t}{m}\right) & otherwise \end{cases} \quad (1)$$

#### 2) Jaro Winkler

Similar to jaro which measures the similarity between two strings by considering the matching characters and the number of transpositions required. This method proposed in 1990 by William E. Winkler gives higher weight to the similarity in the initial characters. This technique is considered better at handling errors in words with similarities at the beginning of the string [5], [6]. The formula of Jaro Winkler is shown in (2).

$$S_{JW}(S1, S2) = S_J(S1, S2) + l \,.\, p \,.\, (1 - S_J(S1, S2)) \quad (2)$$

#### 3) Levenshtein

This method calculates the minimum number of operations (insertions, deletions, and character substitutions) required to change one word into another. Levenshtein is one of the popular methods for normalization because of its ability to correct errors with varying edit distances [6]. The formula of Levenshtein is shown in (3) [7].

$$LD(s,t)_{i,j} = \begin{cases} \max(i,j) & if\ \min(i,j) = 0 \\ min \begin{cases} LD_{(i-1,j)+1} \\ LD_{(i,j-1)+1} \\ LD_{(i-1,j-1)+1_{s_i \neq t_j}} \end{cases} & otherwise \end{cases} \quad (3)$$

The distance between two strings s (source) and t (target) is calculated based on the number of insert, delete, or replace operations required to change s into t. For example, if s is "contooih" and t is "contoh," then LD(s, t) = 2 because it takes two delete operations to remove "o" and "i" from "contooih" to make it "contoh" [8].

#### 4) Damerau Levenshtein

Damerau-Levenshtein is a method similar to Levenshtein but adds a transposition operation as an additional option. This method is more effective than Levenshtein in handling errors caused by adjacent character swaps [9]. For example, if the source string s is "cotnoh" and the target string t is "contoh," then the Levenshtein distance is 2, because two character swap operations are required. However, the Damerau-Levenshtein distance is only 1 because it is sufficient to swap the positions of "n" and "t.". The formula of Levenshtein is shown in (4) [7].

$$DLD(s,t)_{i,j} = \begin{cases} \max(i,j) & if\ \min(i,j) = 0 \\ DLD_{(i-1,j)+1} & if\ i > 0 \\ DLD_{(i,j-1)+1} & if\ j > 0 \\ LD_{(i-1,j-1)+1_{s_i \neq t_j}} & \\ LD_{(i-2,j-2)+1_{s_i \neq t_j}} & if\ i,j > 1\ and\ s_i = t_{j-1}\ and\ s_{i-1} = t_j \end{cases} \quad (4)$$

#### 5) Smith Waterman

This algorithm compares the sequence of two strings to find the optimal local alignment by ignoring mismatched prefixes and suffixes. Since the data has been converted to root words, Smith-Waterman focuses only on character similarities in the middle of the words, giving more head start to small variations or spelling mistakes in the root words [9]. The formula of Levenshtein is shown in (5) [10].

$$H(i,j) = max \begin{cases} 0 \\ H(i-1, j-1) + S/D \\ H(i-1, j) - d \\ H(i, j-1) - d \end{cases} \quad (5)$$

### B. TF-IDF

Term Frequency-Inverse Document Frequency (TF-IDF) is a feature extraction method that assigns a value or weight to each word in a document based on its frequency of occurrence and its uniqueness in the entire document. This method is used to transform text information into a format that can be processed by machine learning algorithms [1], [11], [12], [13].

### C. Lexicon Based Features

Lexicon-Based Features are widely used in sentiment analysis for their practicality and effectiveness. They rely on a sentiment dictionary to match words in text with predefined positive, negative, or neutral sentiments. By combining the sentiment scores of words, this method determines the overall polarity of a text based on its semantic orientation. [1], [14].

### D. Hyperparameter Tuning

The Grey Wolf Optimizer (GWO) is effective in solving optimization problems by mimicking the social hierarchy and hunting strategies of grey wolves. It excels in both exploration and exploitation of the search space, avoiding local optima traps. GWO has shown competitive results against algorithms like PSO and DE, and its adaptive parameter A allocates iterations between exploration and exploitation, enhancing its ability to avoid local minima in complex search spaces. [15].

### E. Support Vector Machine

Support Vector Machine (SVM) is a machine learning method for classification, both on linear and non-linear data, by mapping data to a higher dimension to find the best hyperplane as a separator between classes. This method works by maximizing the margin or distance between two classes, with the hyperplane functioning as the optimal boundary between two data classes [12], [13], [16], [17].

## III. Proposed Method

In this study, we compare language normalization methods, especially spell checking, which corrects words to match words in KBBI, which is expected to improve the accuracy of Twitter sentiment analysis. To address the challenges posed by non-standard language on Indonesian social media, the proposed method was tailored specifically to handle high variability in typographical errors and informal expressions. The normalization process integrates rule-based corrections for common word errors such as insertions, deletions, and transpositions. Additionally, the use of edit-distance algorithms ensures that the normalization method is computationally efficient and flexible, allowing for adaptation to a wide range of error patterns commonly found in Indonesian Twitter data. The results of normalization are then used as input for sentiment analysis to evaluate its effectiveness in improving classification accuracy. This tailored approach aims to bridge the gap between informal language usage and the requirements of NLP systems, enabling a more robust analysis of social media sentiment. The flow of sentiment analysis classification is depicted in Fig. 1.

### *A. Data collection*

This study utilizes sentiment data about the DANA application, sourced from Kaggle, consisting of 50,000 labeled entries [18]. To expedite the analysis, a subset of 500 entries each for positive, negative, and neutral labels was initially used. However, due to low accuracy, the study was revised to include 1,000 positive and 1,000 negative labeled entries. This curated subset preserves the original dataset's sentiment distribution and linguistic diversity, ensuring its representativeness for Indonesian social media contexts.

The accuracy of language normalization methods was tested on 1,000 modified words derived from the KBBI dataset on GitHub, containing 112,651 entries [19]. The test data includes five error types: 1,000 words each with one letter deleted, a random letter added, one letter duplicated, two letters swapped, and one letter replaced. These errors mimic common text variations, as illustrated in Table I. The normalization techniques, based on scalable edit-distance methods, are robust and capable of handling diverse textual errors effectively.

### *B. Preprocessing Data*

Before the data enters the normalization process, each word is preprocessed first through the following steps: converting the words to lowercase (case folding), cleaning symbols, URLs, mentions, and punctuation (data cleaning), removing stopwords (stopword removal), and finally reducing words to their root form (stemming).

### *C. Language Normalization*

In this study, language normalization is applied using five techniques: Jaro, Jaro-Winkler, Levenshtein, Damerau-Levenshtein, and Smith-Waterman. Each technique is applied alternately to obtain accuracy in the language normalization process. The results of each technique will be compared to

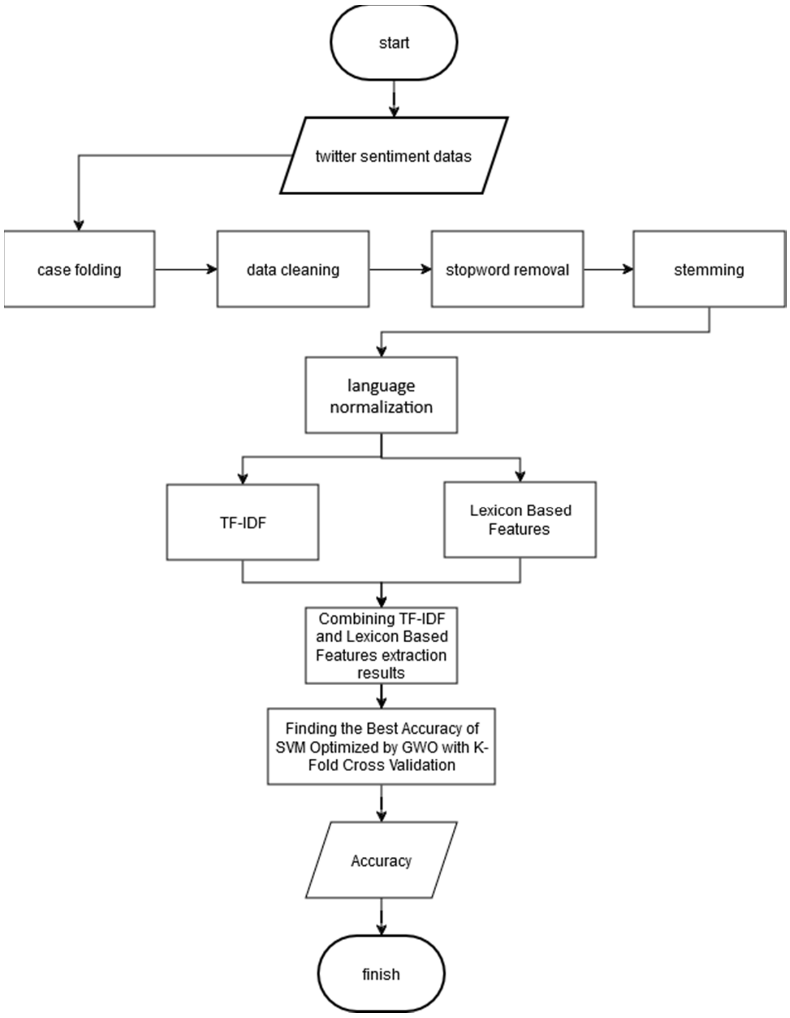


Fig. 1. Proposed Method

TABLE I. EXAMPLE OF MISSPELLING WORD

| No | Error Type | Correct Word | Modified Word |
|---|---|---|---|
| 1 | Delete | makan | mkan |
| 2 | Duplicate | mahkamah | mahkaaamah |
| 3 | Insert | manipulatif | manipulartif |
| 4 | Replace | langsung | aangsung |
| 5 | Swap | gulita | uglita |

determine the most effective approach in improving the accuracy of sentiment analysis.

### D. Sentiment Analysis

Each normalized sentiment is classified using SVM to see the effect of language normalization on sentiment accuracy. Testing is done by measuring the accuracy of sentiment classification on the normalized dataset.

### E. Hyperparameter Tuning

This study uses Grey Wolf Optimizer (GWO) for hyperparameter tuning on Support Vector Machine (SVM) because of its adaptive advantages in handling complex optimization problems. GWO has the ability to explore widely in the search space and then focus on exploitation in promising areas, so it can avoid the trap of local optima.

### F. Evaluation

The evaluation process assesses the impact of language normalization on sentiment analysis accuracy using 10-fold cross-validation (10 CV) for reliable results. Each fold is tested to avoid overfitting and understand the overall performance of the model. In addition, Grey Wolf Optimizer (GWO) is used for SVM hyperparameter tuning with 10 CV, providing insight into the contribution of GWO in improving SVM accuracy on the normalized dataset and its effectiveness in handling data variations.

## IV. RESULTS AND DISCUSSIONS

After conducting the experiment, the following results were obtained:

### A. Language Normalization

Based on Fig. 2. it can be seen that:

#### 1) Jaro

Jaro's method has a high score in duplicate and insert detection, which is 95.3%, indicating its reliability in recognizing these types of changes. However, its performance is low in detecting replaces (57.7%), indicating its weakness in detecting character changes with accuracy. Overall, the average score of 81.3% shows that Jaro performs quite well on various changes, except for replaces.

#### 2) Jaro Winkler

Jaro Winkler performed better than Jaro, especially in detecting duplicates (96.6%) and inserts (95.7%), and was slightly stronger in detecting replaces (60.3%). The overall average of 81.76% shows that Jaro Winkler outperforms Jaro in almost all aspects. Jaro Winkler's superiority is likely due to its emphasis on the initial characters of the string, which helps identify changes in words that have similarities at the beginning.

#### 3) Levenshtein

The Levenshtein method is quite strong in detecting inserts (95.5%) and replaces (76.8%), but weak in detecting deletes (51.8%) and swaps (36.8%). The average score of 69.58% indicates that this method is effective for basic changes such as inserts and replaces, but less than optimal for deletes and swaps.

Levenshtein also does not consider the relative position of characters, making it less effective in detecting positional changes such as swaps.

#### 4) Damerau Levenshtein

This method is able to detect insert changes well (95.5%) and has a higher score on swaps (83.2%) compared to the Levenshtein method. The average score of 78.78% shows the superiority of Damerau-Levenshtein in handling character position changes. This performance makes it effective for applications involving typos, especially those involving character position changes.

#### 5) Smith Waterman

Smith Waterman has a lower average score of 53.02%, and does not excel at any particular type of change, likely due to its focus on local substring matching. This method is effective for finding similarities in specific parts of text, but is less than optimal for detecting changes that span the entire text.

### B. Sentiment Analysis

After preprocessing, data normalization using various methods significantly improved sentiment analysis accuracy, as shown in Table II. Without normalization, accuracy was only 48.57%, highlighting how unrecognized word variations reduce model performance. Normalization boosted accuracy across all methods, with Jaro leading at 51.56% due to its strength in handling duplication and insertion errors, followed by Jaro Winkler at 51.36%. Levenshtein and Damerau-Levenshtein reached around 49.5%, with the latter offering a slight edge by managing character swaps. Smith-Waterman improved accuracy to 50.27%, leveraging its ability to detect

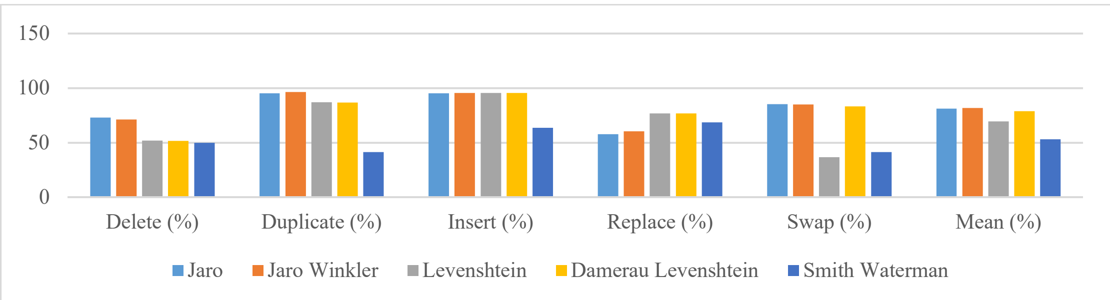


Fig. 2. Accuracy of Language Normalization Method

TABLE II. SENTIMENT ANALYSIS ACCURACY

| Language Normalization Method | Sentiment Analysis Accuracy |
|---|---|
| Without Languange Normalization | 48.57 % |
| Jaro | 51.56 % |
| Jaro Winkler | 51.36 % |
| Levenshtein | 49.4949% |
| Damerau Levenshtein | 49.4953% |
| Smith Waterman | 50.27 % |

character similarities in longer words. Despite these gains, the accuracy remains relatively low, suggesting the need for further optimization, such as applying Grey Wolf Optimizer (GWO) to enhance SVM's performance in leveraging normalization techniques.

### *C. Hyperparameter Tuning with GWO*

Hyperparameter tuning with the Grey Wolf Optimization (GWO) method significantly improved the accuracy of sentiment analysis using the RBF and Polynomial kernels, as shown in Table III. The Smith Waterman method achieved the highest accuracy of 68.8% with the Polynomial kernel, attributed to the kernel's ability to handle complex relationships in data, effectively addressing language variations. Jaro demonstrated strong performance with 68.25% accuracy using the RBF kernel, likely due to its capability to capture non-linear patterns matching minor character errors such as duplication and insertion. Meanwhile, Jaro Winkler achieved 67.84% accuracy with the Polynomial kernel, close to Jaro's performance on RBF, highlighting the Polynomial kernel's effectiveness in capturing language variations with a focus on initial character changes. Levenshtein and Damerau-Levenshtein achieved accuracies of 67.58% and 66.91% with Polynomial and RBF kernels, respectively; despite improvement after tuning, their performance was lower than methods like Jaro and Smith Waterman, suggesting their limitations in capturing complex patterns. These results underscore the importance of selecting optimal kernels, as highlighted by previous studies [20], which emphasize that the choice of kernel significantly impacts the classification performance in SVM. Optimal kernel selection and parameter tuning are critical for sentiment analysis, as the choice of kernel and hyperparameters depends on the data, requiring careful tuning for the best results.

After hyperparameter tuning, the classification accuracy remained low, especially in the neutral class in the three-class sentiment analysis. Therefore, the analysis was repeated using 2000 data with two classes (1000 positive and 1000 negative), which significantly improved the model accuracy. The Damerau-Levenshtein method achieved the highest accuracy of 89.15% (Table IV), while Jaro recorded 68.25% (RBF kernel) on three classes (Table III) and 88.89% on two classes, slightly below the unnormalized data (88.8949%) but higher than Jaro-Winkler (88.59%). This study confirms the importance of language normalization in informal text sentiment analysis, with Damerau-Levenshtein excelling in complex error correction, while Jaro is consistent in multi-class scenarios. These findings indicate that the success of a method depends not only on error correction but also on its suitability to the needs of sentiment analysis. As shown in Fig. 3, accuracy improves for each normalization method after hyperparameter tuning and the removal of the neutral class.

TABLE III. HYPERPARAMETER TUNING RESULTS

| Language Normalization Method | Best Kernel | Best C | Best Gamma | Accuracy (%) |
|---|---|---|---|---|
| Without Languange Normalization | Polynoamial | 61.74 | 1.00 | 66.65 |
| Jaro | RBF | 1.44 | 0.54 | 68.25 |
| Jaro Winkler | Polynomial | 43.40 | 0.98 | 67.84 |
| Levenshtein | Polynomial | 0.88 | 0.94 | 67.58 |
| Damerau Levenshtein | RBF | 1.54 | 0.46 | 66.91 |
| Smith Waterman | Polynomial | 2.33 | 0.24 | 68.80 |

TABEL IV. HYPERPARAMETER TUNING RESULTS WITHOUT NETRAL LABEL

| Language Normalization Method | Best Kernel | Best C | Best Gamma | Accuracy |
|---|---|---|---|---|
| Without Languange Normalization | RBF | 0.63 | 1.00 | 88.8949 % |
| Jaro | RBF | 1.28 | 0.96 | 88.8944 % |
| Jaro Winkler | Polynomial | 1.93 | 1.00 | 88.59 % |
| Levenshtein | Polynomial | 0.67 | 0.77 | 89.05 % |
| Damerau Levenshtein | Polynomial | 0.59 | 0.61 | 89.15 % |
| Smith Waterman | RBF | 0.42 | 1.00 | 87.80 % |

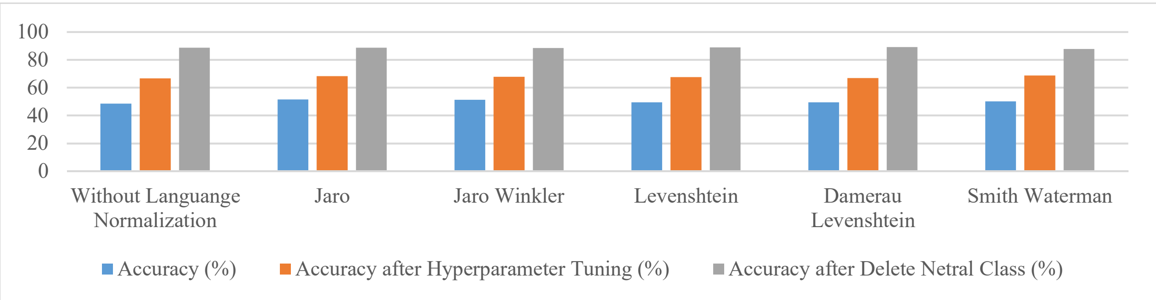


Fig. 3. Comparison of Sentiment Analysis Accuracy for Each Process

## V. CONCLUCION

The use of informal language on Indonesian social media, particularly on platforms like Twitter, presents significant challenges in sentiment analysis due to its diverse and inconsistent nature. This study evaluates edit-based text distance algorithms, including Jaro, Jaro-Winkler, Levenshtein, Damerau-Levenshtein, and Smith-Waterman, for their effectiveness in correcting spelling errors. These methods are practical and efficient, as they address text variations directly using predefined rules and can adapt to a wide range of typographical variations, avoiding the limitations of AI-based models that require constant retraining to handle new errors and emerging words.

Experimental results show that the Damerau-Levenshtein method achieves the highest accuracy of 89.15% in two-class sentiment classification (positive and negative), outperforming other methods. Jaro demonstrated robust performance in handling character duplication and insertion, contributing to consistent sentiment analysis results, albeit slightly below Damerau-Levenshtein. Additionally, Jaro consistently delivers competitive results compared to other language normalization methods, making it a reliable option for various applications. This consistency, combined with its robustness, positions Jaro as a strong candidate for consideration in real-world sentiment analysis tasks. The ability of edit-based methods to handle extensive error combinations without retraining further underscores their effectiveness and versatility.

Future research should focus on developing normalization techniques capable of managing more complex errors and incorporating regional language variations in Indonesia. Additionally, integrating advanced machine learning techniques, such as deep learning with word embeddings, could further enhance the model's ability to process informal text and improve sentiment analysis accuracy in NLP applications.

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